

CSA15 Cloud Computing and Big Data Analytics

CO3 AT1 - Guided Cloud Access Experiments Workbook

Course Outcome	CO3 - Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications.
Unit Alignment	Unit 3 - Access to Cloud: cloud access, platforms, web applications, APIs, storage models, SAN, serverless, edge/fog, collaboration, security and development workflow.
Faculty	Dr C. Rajesh Babu, Professor, Department of CSE
Employee ID	SSETSCS415
Submission Mode	Individual PDF evidence through Google Classroom and GitHub repository link.

How This Workbook Works

Stage	Student Action
Learn	Read the short technical note before touching the tool.
Do	Perform the guided action using the recommended tool.
Observe	Check what changed on the screen or in the output.
Record	Paste screenshots, URLs, tables, command outputs or diagram references.
Explain	Write the technical reasoning in your own words.
Extend	Apply the same concept to the capstone product continued from CO2 AT3.
Submit	Upload the completed evidence to GitHub and Google Classroom.

Important Safety Rule

Screenshots must not reveal passwords, private keys, API tokens, payment details, personal account recovery information or sensitive user data. Crop or mask sensitive values before submission.

Student and Faculty Details

Student Name	Vallapu Mohan Kumar	Register Number	192524225
Class / Section	B.TECH2 ND YEAR (AI&DS)	Slot	BC
Course	CSA15 Cloud Computing and Big Data Analytics	Assessment	CO3 AT1
Capstone Title	AGRIMARKET AI:CROP PRICE FORECAST AND SELLING DECISION APP	Submission Date	22-08-2026
Faculty	Dr C. Rajesh Babu	Employee ID	SSETSCS415
Department	Computer Science and Engineering	GitHub Link	

Tool Links

Tool	Purpose	Link
Firebase Console	Cloud storage and project evidence	https://console.firebase.google.com/
Postman	REST API request-response evidence	https://www.postman.com/downloads/
JSONPlaceholder	Safe public API for GET/POST practice	https://jsonplaceholder.typicode.com/
ReqRes	Safe public API for request-response practice	https://reqres.in/
draw.io	Architecture and workflow diagrams	https://app.diagrams.net/
GitHub	Repository evidence submission	https://github.com/
GitHub Pages	Simple hosted web access evidence	https://pages.github.com/

Experiment 1: Cloud Storage Model Comparison and Selection

Original CO3 Question	Create and document a cloud storage comparison between object, block, and file storage. Select the best storage model for backups, VM disks, and shared enterprise documents.
Tools	Firebase Console, Firebase Storage, draw.io, browser, GitHub.

CO2 AT3 Continuity

Use the same capstone title and infrastructure direction already defended in CO2 AT3. Do not repeat VM sizing; focus on cloud service access.

Learn

A cloud application may store files, database records, VM disks, images, logs and shared documents. Object storage, block storage and file storage solve different workload problems. Firebase Storage is used here as a beginner-friendly object storage console because students can create a project, open a storage bucket, upload a file, observe metadata and reason about access control from the browser.

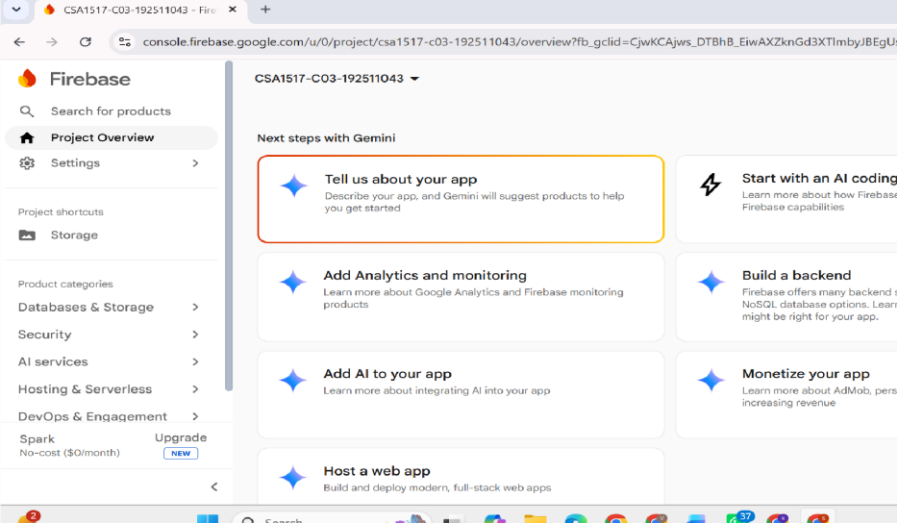
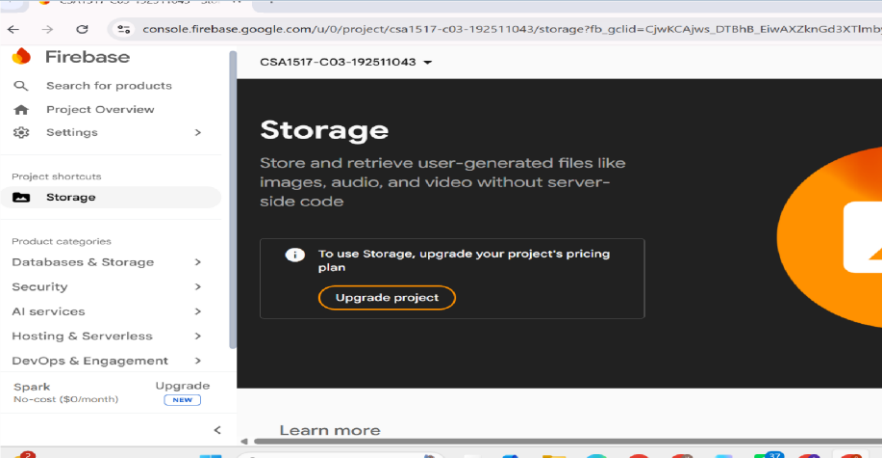
Do

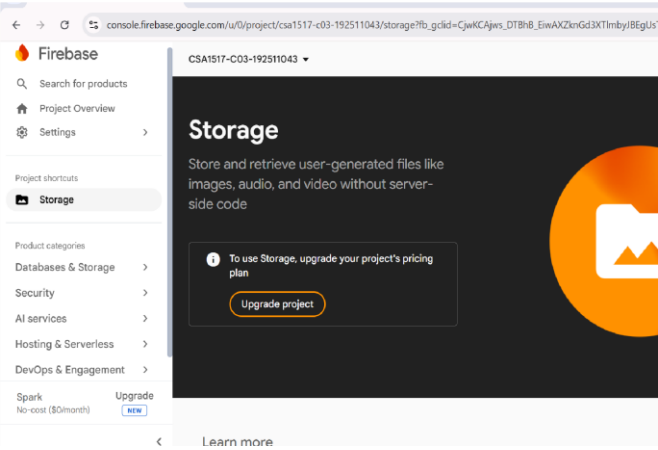
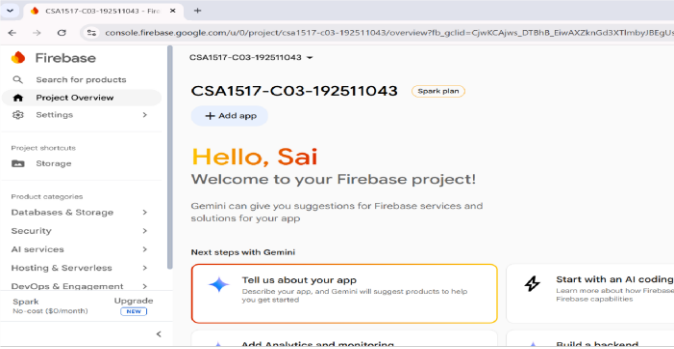
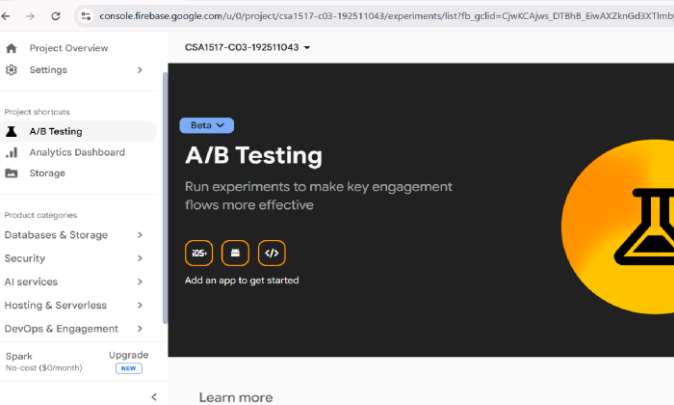
- Begin in the browser at <https://console.firebase.google.com/>. Sign in with the account approved for academic work. After the console opens, use Add project or Create a project. Give the project a course-friendly name such as csa15-co3-regno. If Google Analytics is shown, keep it disabled unless faculty instructs otherwise. Create the project and wait until the project dashboard opens. The screen to notice is the Firebase project overview page.
- In the left navigation area, open Build and choose Storage. If Storage is not yet enabled, choose the option to get started. Select the default bucket location shown by Firebase or the nearest available region. Use test access only for lab observation if the console requires an initial rule choice, and remember that production apps must use strict authenticated rules. The screen to notice is the storage bucket page.
- Create a small text file on the computer named student_backup.txt. Add two lines inside it: the student register number and the capstone title. Return to Firebase Storage and use Upload file. Select student_backup.txt and wait until it appears in the bucket list. The screen to notice is the object list showing the uploaded file.
- Open the uploaded file entry. Observe the metadata area: file name, file size, content type, storage path, creation/update time and access status. If a download URL or access token is visible, do not paste sensitive token values in the workbook. Record only the safe evidence: filename, type, size and storage path.
- Open draw.io and draw a three-part comparison diagram: object storage for uploaded files/backups, block storage for VM disks, and file storage for shared folders. Under the diagram, write where your capstone will use storage and why Firebase Storage represents object storage in this experiment.

Observe

Check whether the uploaded file is visible, whether metadata is displayed, and whether public/private access is controlled. Do not expose sensitive files.

Record Evidence

Evidence Item	Student Reference / Screenshot Number	Checkpoint
Cloud storage dashboard screenshot.		[] Completed
Uploaded object/file screenshot.		[] Completed

Metadata screenshot.		[] Completed
Completed storage model comparison table.		[] Completed
Capstone storage recommendation table.		[] Completed

Explain

Why is one storage model not suitable for all workloads? Justify using backup, VM disk and shared-folder examples, then extend the answer to your capstone product.

Student Explanation Space

Why is one storage model not suitable for all workloads?

Different storage models are designed for different purposes, so a single storage model cannot efficiently support every workload.

- **Backup Storage (Object Storage):** Best for storing large amounts of backup data, images, videos, and archives. It is cost-effective and highly scalable but not suitable for running operating systems or databases that require fast disk access.
- **VM Disk Storage (Block Storage):** Provides high-speed read and write operations required by virtual machines and databases. However, it is more expensive and not ideal for storing large backup files.
- **Shared Folder Storage (File Storage):** Allows multiple users and applications to access the same files through a common folder structure. It is useful for collaboration but may not provide the performance needed for databases or VM disks.

Extension to Capstone Product (AgriMarket AI Crop Price Forecast and Selling Decision App):

The AgriMarket application requires multiple storage types:

- **Object Storage** for storing uploaded crop images, reports, and historical datasets.
- **Block Storage** for hosting virtual machines running AI models and application services.
- **File Storage** for sharing datasets, reports, and project files among developers and administrators.

Therefore, using a combination of storage models improves performance, scalability, cost efficiency, and reliability. No single storage model can meet all the requirements of backups, virtual machines, shared access, and AI-based agricultural analytics.

Extend to Capstone

Select the storage model required for your capstone product: user uploads, app images, database attachments, logs, backups or shared documents.

Capstone Title	AGRIMARKET AI:CROP PRICE FORECAST AND SELLING DECISION APP
CO2 AT3 Infrastructure Decision Carried Forward	Use cloud storage for scalable and reliable data management in the Agri Market application.

Cloud Service / Access Layer Added in this Experiment	Firestore Storage (Object Storage) for storing uploaded files, crop images, reports, and backup data.
Risk or Limitation Identified	Test mode storage rules may allow unauthorized access. Large file uploads can increase storage costs
Improvement for Prototype or Pilot	Implement authenticated access rules, enable monitoring, use automatic backups, and optimize file organization for better security and performance.

Submit

GitHub Folder	CO3_AT1/Experiment_1
Files to Upload	Completed PDF, diagrams, screenshots and any exported table.
Google Classroom Entry	Paste GitHub repository link and attach final PDF.

Experiment 2: REST API Workflow for a Cloud-Based Student Portal

Original CO3 Question	Develop a simple REST API workflow for a cloud-based student portal. Show request, response, authentication, and deployment flow.
Tools	Postman Web/Desktop, JSONPlaceholder or ReqRes, draw.io, browser, GitHub.
CO2 AT3 Continuity	Use the same capstone title and infrastructure direction already defended in CO2 AT3. Do not repeat VM sizing; focus on cloud service access.

Learn

REST API means Representational State Transfer Application Programming Interface. For beginners, think of it as a controlled doorway through which a client app asks a cloud service for data or sends data to be processed. A browser or mobile app sends a request to an API endpoint. The API validates the request, applies business logic, communicates with storage or database services, and returns a structured response. Postman is used because it lets students send requests, view status codes, inspect headers, read JSON responses and test API behavior without writing a full app first.

Do

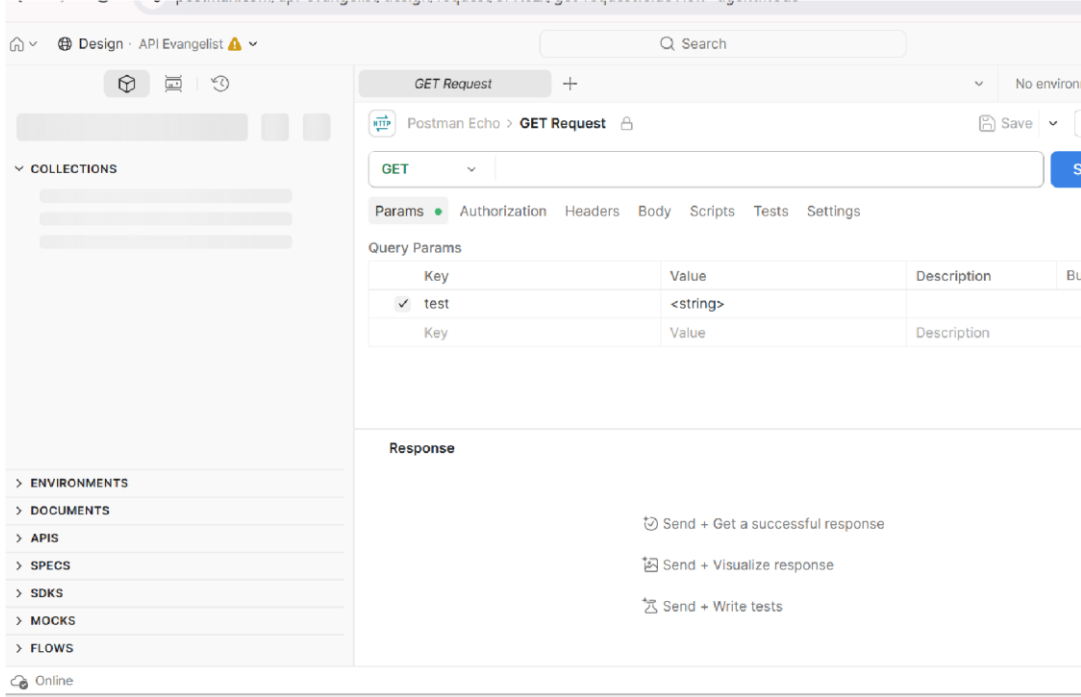
- Open <https://www.postman.com/> or the Postman desktop application. Sign in or create a free account if Postman asks for one. After the workspace opens, create a personal workspace or use the default workspace. Create a collection named CSA15_CO3_AT1_API so the evidence is organized.

- Open a new request tab. Keep the method as GET. Enter <https://jsonplaceholder.typicode.com/users> in the request URL field and send the request. Look at the response area. The expected observation is a successful status such as 200 OK and a JSON response containing user-like records.
- Read one object from the JSON response. Identify fields such as id, name, username, email, address and company. Record three fields in the workbook and explain what each field would mean in a student portal or capstone product.
- Create another request with method POST. Use <https://jsonplaceholder.typicode.com/posts> as the URL. Open the Body area, choose raw, select JSON, and enter a small JSON body such as `{"title": "CSA15 cloud test", "body": "API evidence", "userId": 1}`. Send the request. The expected observation is a created/test response, often with status 201 or a simulated response.
- Open draw.io and draw the request-response flow. Show Browser/Mobile Client -> REST API Endpoint -> Backend Logic > Database/Storage -> JSON Response. Add a small authentication gate before backend logic to show that real cloud applications should protect API access.

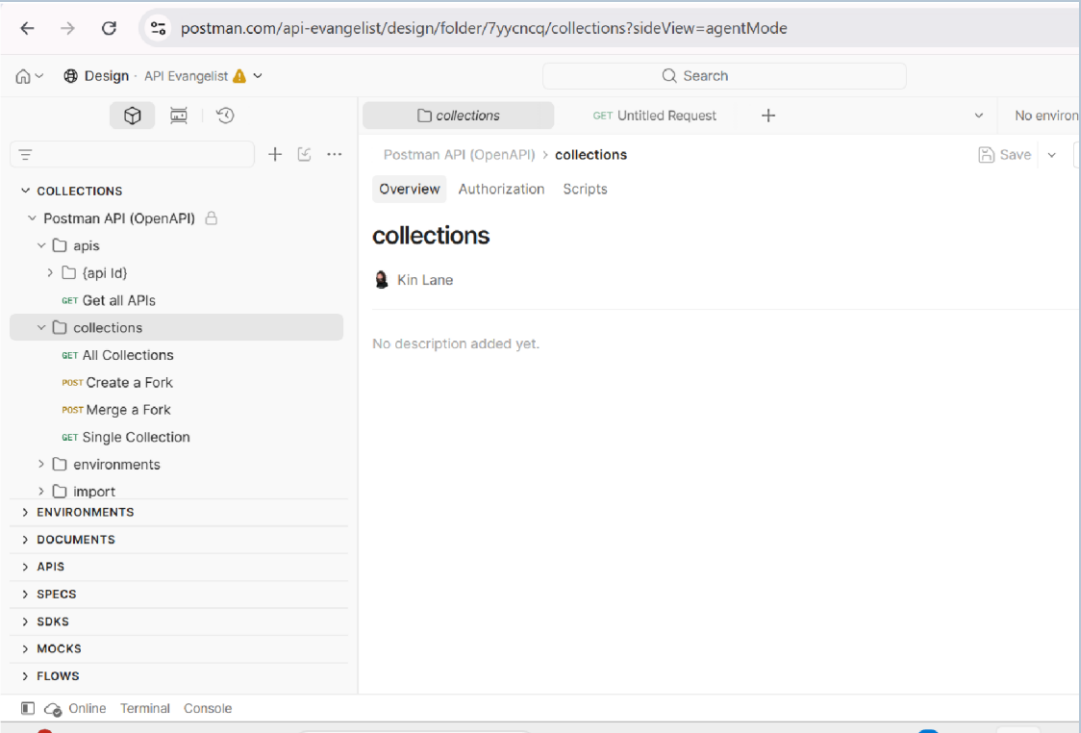
Observe

Verify whether the request returns status 200 or a valid test response. Identify JSON fields and explain what each field represents.

Record Evidence

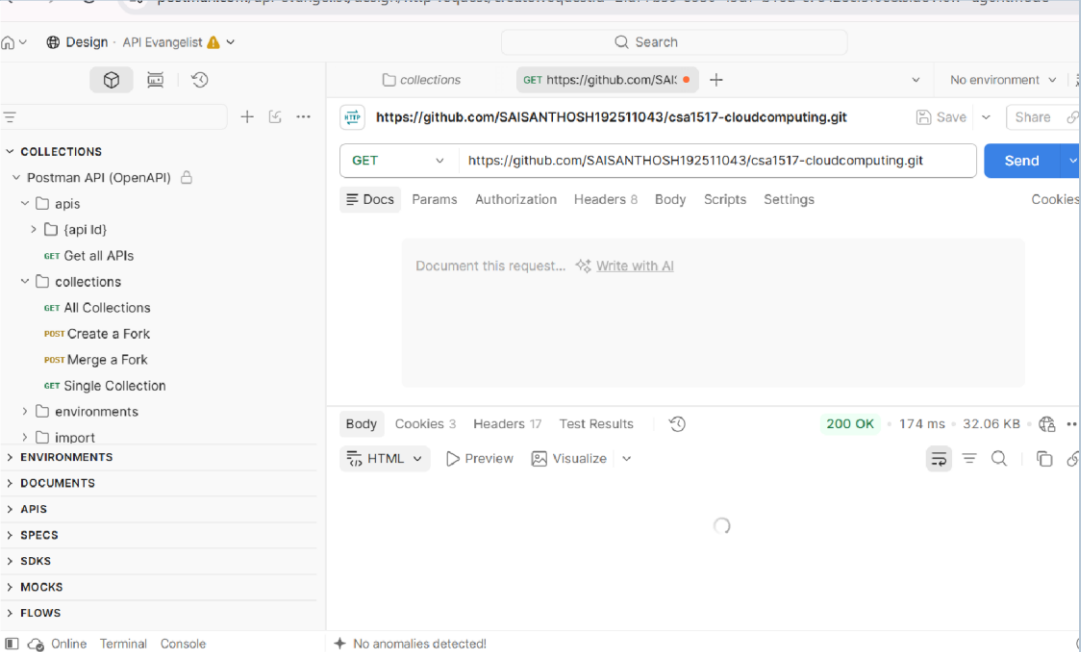
Evidence Item	Student Reference / Screenshot Number	Checkpoint
Postman GET request screenshot.		[] Completed

Status code and response time screenshot.

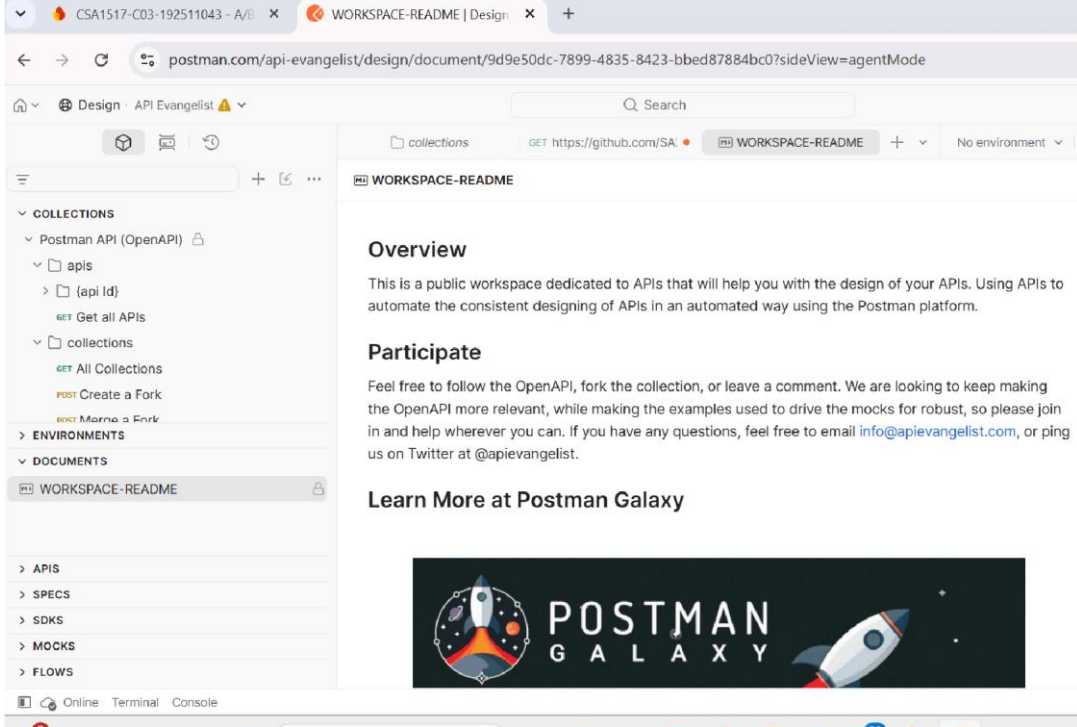
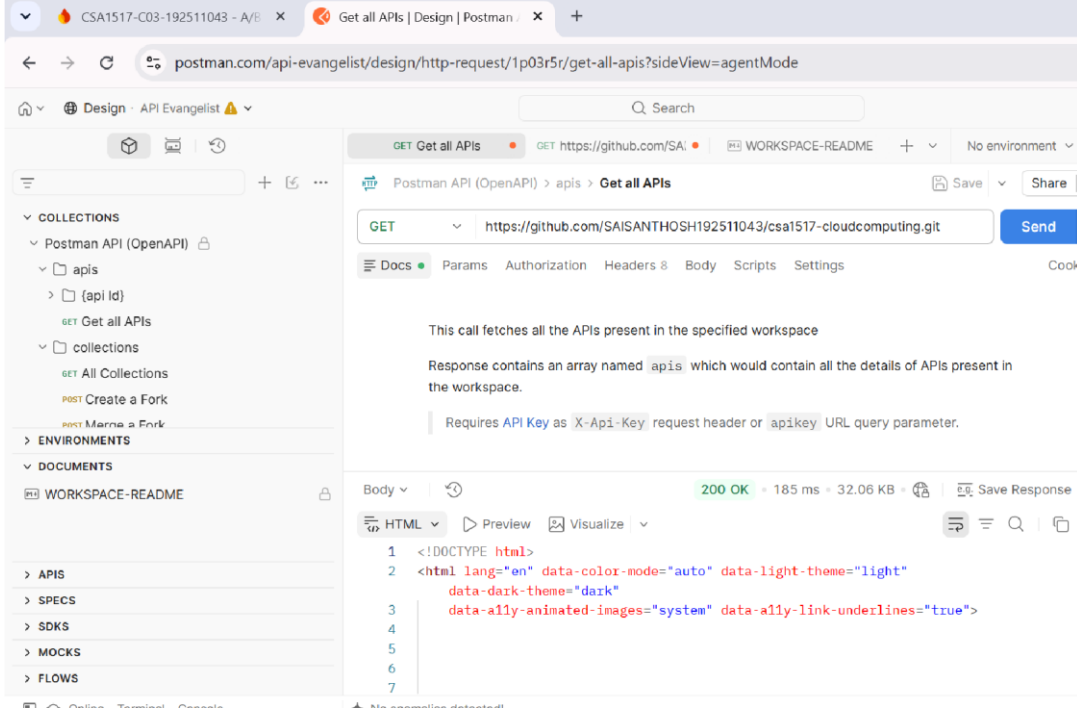


[] Completed

JSON field identification table.



[] Completed

POST request body and response screenshot.		[] Completed
API workflow diagram.		[] Completed

Explain

Why should a cloud application expose data through APIs instead of allowing users to directly access the database?

Student Explanation Space

student Explanation

Cloud applications should expose data through **APIs** instead of allowing users to directly access the database because APIs provide **security, control, and scalability**.

- **Security:** APIs authenticate and authorize users before granting access to data, preventing unauthorized access.
- **Data Validation:** APIs check and validate input data before it is stored or retrieved from the database.
- **Business Logic:** APIs enforce application rules and processing logic, ensuring data consistency.
- **Scalability:** APIs can handle many client requests efficiently without exposing the database to heavy traffic.
- **Maintainability:** Changes to the database structure can be made without affecting users, as they interact only with the API.

Capstone Example (AgriMarket AI Crop Price Forecast and Selling Decision App):

Farmers use the mobile application to request crop price predictions. The request is sent to an API, which verifies the user, processes the request, retrieves data from the database, and returns the prediction. Direct database access would be insecure and could expose sensitive user and market data.

Extend to Capstone

Write two API endpoints for your capstone product and explain the input, backend action, storage access and response.

Capstone Title	AGRIMARKET AI:CROP PRICE FORECAST AND SELLING DECISION APP
CO2 AT3 Infrastructure Decision Carried Forward	Use a secure API-based architecture between users and cloud databases.
Cloud Service / Access Layer Added in this Experiment	REST API layer tested using Postman (GET and POST requests).
Risk or Limitation Identified	APIs without authentication can expose sensitive data and may be vulnerable to misuse or attacks.
Improvement for Prototype or Pilot	Add API authentication, authorization, rate limiting, input validation, and HTTPS encryption for secure access.

Submit

GitHub Folder	CO3_AT1/Experiment_2
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Files to Upload	Completed PDF, diagrams, screenshots and any exported table.
Google Classroom Entry	Paste GitHub repository link and attach final PDF.

Experiment 3: Serverless Event Flow for Image Upload Processing

Original CO3 Question	Design a serverless event flow for image upload processing using object storage, function-as-a-service, and notification service.
Tools	draw.io, Firebase Storage evidence from Experiment 1, optional Pipedream/Make simulation, GitHub.
CO2 AT3 Continuity	Use the same capstone title and infrastructure direction already defended in CO2 AT3. Do not repeat VM sizing; focus on cloud service access.

Learn

Serverless computing allows a cloud platform to run code in response to events without the student managing the server directly. In this experiment, students do not need paid cloud function deployment. They design and document a correct event-driven flow. The key idea is: an event happens, a cloud function is triggered, the function performs a small task, and another cloud service records or sends the output.

Do

- Start with the uploaded file evidence from Experiment 1. Treat the upload as the event. For capstone mapping, choose one event such as image upload, booking creation, complaint submission, attendance scan, payment update or alert generation.
- In the workbook, write the event source in a short table. Mention who performs the action, what data is created, where it is stored, and what should happen automatically after the event.
- Design the cloud function behavior without writing production code. Name the function clearly, for example processUploadedImage, confirmBooking, validateAttendance or sendIssueAlert. Write what the function receives, what it checks, what it changes, and what output it produces.
- Decide the output service. It may be a notification, email, database status update, dashboard row, resized image, log entry or report update. Also decide what should happen if the function fails: retry, error log, faculty/admin alert or manual review.
- Draw the event flow in draw.io using this visual order: User Action -> Cloud Storage/Database Event -> Serverless Function -> Processing Decision -> Output Service -> Log/Monitoring. Paste the diagram and complete the trigger-actionoutput table.

Observe

Identify the trigger, processing function, output service, failure point and retry/alert requirement. If billing blocks real function deployment, submit a technically accurate simulation and architecture evidence.

Record Evidence

Evidence Item	Student Reference / Screenshot Number	Checkpoint
Event source evidence or simulated event table.	Farmer uploads crop image	[] Completed
Serverless flow diagram.	Farmer/User	[] Completed
Trigger-action-output table.	Crop image file and upload details	[] Completed
Failure and retry note.	Firebase Storage (Object Storage)	[] Completed
Capstone event mapping.	Trigger image analysis and update prediction results	[] Completed

Explain

Why is serverless suitable for upload/event processing? Mention one limitation also.

Student Explanation Space

Serverless computing is suitable for upload and event processing because it automatically runs code whenever an event occurs, such as a file upload, database update, or form submission. It scales automatically, reduces infrastructure management, and charges only for actual usage, making it costeffective for cloud applications.

Example: When a farmer uploads a crop image, a serverless function can automatically validate the image, analyze it, update the database, and send a notification without requiring a continuously running server.

One

Limitation:

Serverless functions may experience **cold start delays**, where the function takes slightly longer to execute if it has not been used recently. This can affect applications that require immediate response times

Extend to Capstone

Apply the serverless idea to one real capstone workflow, such as image processing, booking confirmation, attendance validation, alert generation or report update.

Capstone Title	AGRIMARKET AI:CROP PRICE FORECAST AND SELLING DECISION APP
CO2 AT3 Infrastructure Decision Carried Forward	Use a secure API-based architecture between users and cloud databases
Cloud Service / Access Layer Added in this Experiment	Firebase Storage (Object Storage) for storing uploaded files, crop images, reports, and backup data.

Risk or Limitation Identified	APIs without authentication can expose sensitive data and may be vulnerable to misuse or attacks.
Improvement for Prototype or Pilot	Add API authentication, authorization, rate limiting, input validation, and HTTPS encryption for secure access.

Submit

GitHub Folder	CO3_AT1/Experiment_3
Files to Upload	Completed PDF, diagrams, screenshots and any exported table.
Google Classroom Entry	Paste GitHub repository link and attach final PDF.

Experiment 4: Edge and Fog Computing for Smart Traffic Management

Original CO3 Question	Compare edge computing and fog computing for a smart traffic management system. Include architecture, latency considerations, and deployment location.
Tools	draw.io, Google Sheets or Excel latency table, optional Wokwi/IoT screenshot only if faculty permits, GitHub.
CO2 AT3 Continuity	Use the same capstone title and infrastructure direction already defended in CO2 AT3. Do not repeat VM sizing; focus on cloud service access.

Learn

Cloud computing is powerful but may be physically distant from sensors and users. Edge computing places small decisions very near the data source. Fog computing introduces an intermediate distributed layer between edge devices and the central cloud. This experiment is an architecture reasoning activity; students are not required to build a real traffic sensor. The correct output is a placement decision supported by latency and data-volume reasoning.

Do

- Use this scenario: traffic cameras and road sensors produce vehicle count, speed, congestion level and emergency vehicle alerts. Some actions must happen immediately at the signal, some can be handled at a regional traffic control unit, and some can be stored in the central cloud for analysis.
- Classify each task into edge, fog or cloud. Put instant signal switching and emergency alert detection near edge. Put junction-level aggregation and area coordination in fog. Put long-term storage, analytics dashboard and city-level reports in cloud.
- Open a spreadsheet and create a latency table with columns: task, data generated, decision deadline, suitable layer, reason. Use example decision deadlines such as 1 second for emergency signal response, 5-10 seconds for regional coordination and minutes/hours for reports. The exact numbers are reasoning aids, not measured network results.

□ Draw the architecture in draw.io. Show sensors/cameras at the road, edge device at signal controller, fog node at regional traffic control, cloud platform for storage/analytics and dashboard for administrator. Use arrows to show data movement and labels to show why each layer is used.

□ Connect the same idea to the capstone. If the capstone has live location, sensor, mobile offline, queue, emergency, QR attendance or fast alert behavior, identify what can be edge/fog/cloud. If the capstone does not need edge/fog, state that browser/mobile client -> cloud service access is sufficient and justify it.

Observe

Check whether time-critical actions are placed closer to data sources and whether long-term storage/analytics remain in the cloud.

Record Evidence

Evidence Item	Student Reference / Screenshot Number	Checkpoint
Task classification table.		[] Completed
Latency reasoning table.		[] Completed
Edge-fog-cloud architecture diagram.		[] Completed
Deployment location justification.		[] Completed
Capstone relevance note.		[] Completed

Explain

Which part of the system should run at edge, fog and cloud? Justify with latency and data-volume reasoning.

Student Explanation Space

student Explanation Space

Which part of the system should run at edge, fog and cloud?

- **Edge Layer:** Runs close to the user device and handles immediate data collection and quick decisions. Examples include mobile phones, sensors, cameras, and attendance scanners. Edge processing reduces latency because data is processed near the source.
- **Fog Layer:** Acts as an intermediate layer between edge devices and the cloud. It performs temporary processing, filtering, aggregation, and routing of data. This reduces network traffic and improves response time.
- **Cloud Layer:** Handles large-scale storage, analytics, machine learning, reporting, backups, and longterm data management. The cloud is suitable for high-volume data processing but has higher latency than edge processing.

Latency Reasoning:

Tasks requiring instant responses should run at the edge or fog layer. Tasks that are not time-critical and require large computational resources should run in the cloud.

Data Volume Reasoning:

Raw sensor data can be filtered at the edge or fog layer before sending only important information to the cloud, reducing bandwidth usage and storage costs.

Extend to Capstone

If your capstone has location-sensitive, sensor-based, offline-first or fast-alert requirements, identify the edge/fog/cloud placement. If not applicable, justify why direct cloud access is sufficient.

Capstone Title	AGRIMARKET AI:CROP PRICE FORECAST AND SELLING DECISION APP
CO2 AT3 Infrastructure Decision Carried Forward	Use a secure API-based architecture between users and cloud databases
Cloud Service / Access Layer Added in this Experiment	Firebase Storage (Object Storage) for storing uploaded files, crop images, reports, and backup data.
Risk or Limitation Identified	APIs without authentication can expose sensitive data and may be vulnerable to misuse or attacks.
Improvement for Prototype or Pilot	Add API authentication, authorization, rate limiting, input validation, and HTTPS encryption for secure access.

Submit

GitHub Folder	CO3_AT1/Experiment_4
Files to Upload	Completed PDF, diagrams, screenshots and any exported table.
Google Classroom Entry	Paste GitHub repository link and attach final PDF.

Experiment 5: Cloud Application Deployment and Access Workflow

Original CO3 Question	Demonstrate a cloud application deployment workflow showing client access, browser interaction, web API call, storage access, and monitoring.
Tools	GitHub Pages as primary beginner path, Firebase Hosting as optional path, Postman, draw.io, GitHub repository, browser.
CO2 AT3 Continuity	Use the same capstone title and infrastructure direction already defended in CO2 AT3. Do not repeat VM sizing; focus on cloud service access.

Learn

A cloud-backed application is accessed through a browser or mobile client. The user action travels through frontend, API/backend, authentication, platform services, storage/database and monitoring/logging components. GitHub Pages is the primary beginner path because it can host a simple static page from a repository. Firebase Hosting may be used when the student is comfortable with Firebase CLI or Cloud Shell.

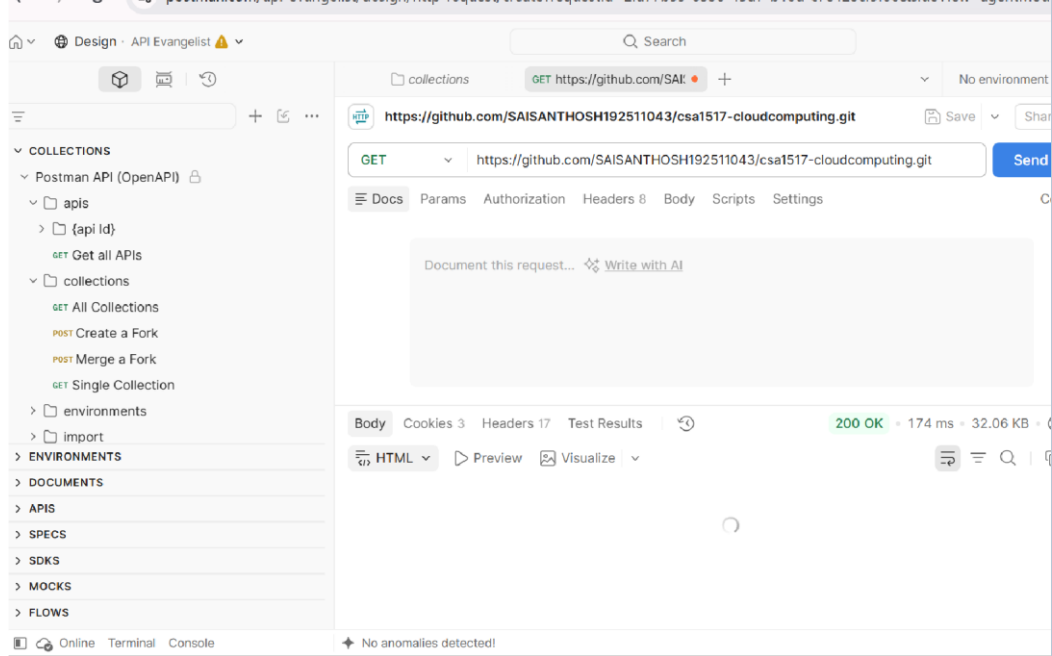
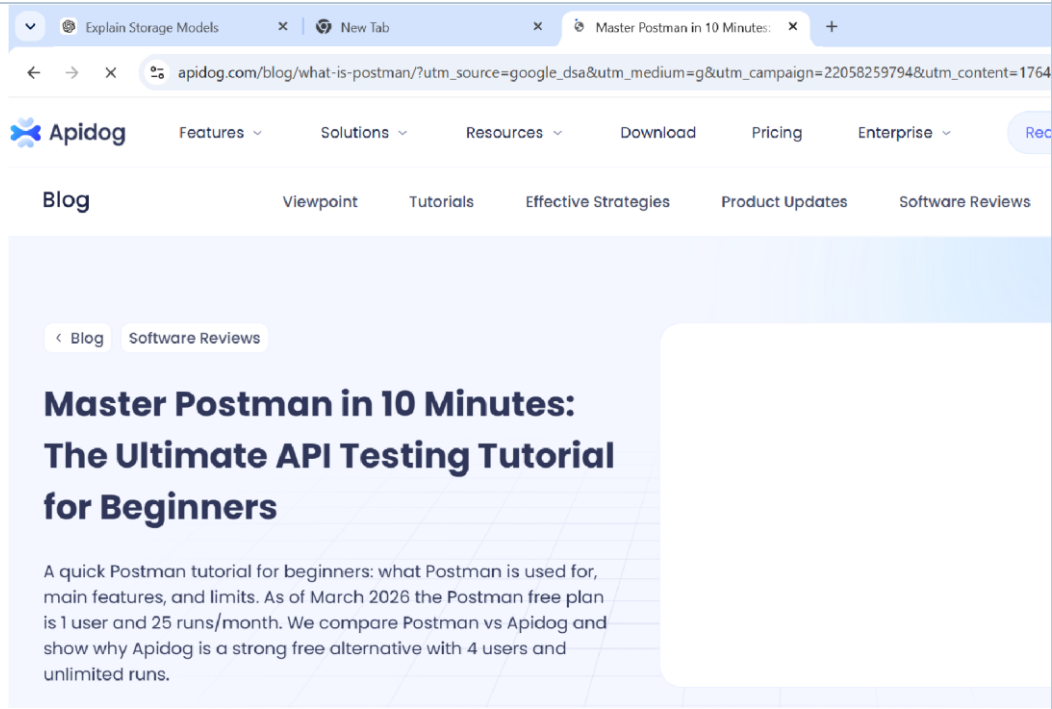
Do

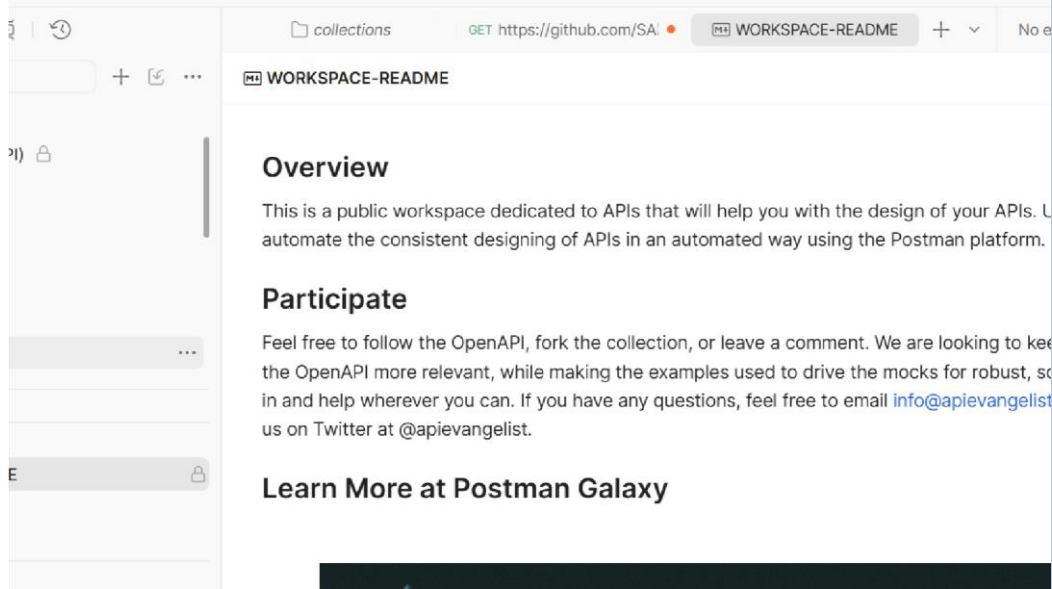
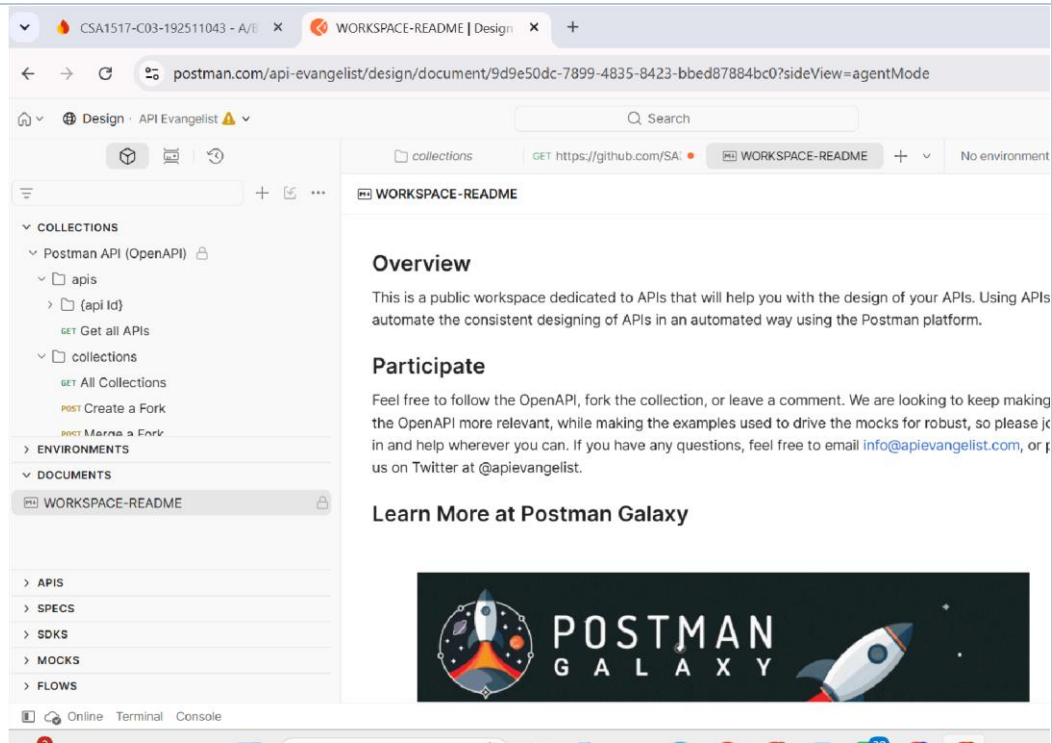
- ☐ Create or open a GitHub repository for CO3 AT1. Add a simple index.html file. The page should show the capstone title, one user action and one cloud service used. Example page sections: product name, user action, API endpoint idea, storage used and evidence link.
- ☐ In the repository settings, open Pages. Select the main branch and root folder as the publishing source when available. Save the Pages setting and wait for GitHub to publish the site. Open the generated Pages URL in the browser and verify that the page loads.
- ☐ If Firebase Hosting is used instead, open Firebase Console, choose Hosting, follow the hosting setup guidance, and deploy using Firebase CLI or Cloud Shell only when faculty confirms that the device and account are ready. For most beginners, GitHub Pages evidence is sufficient for this experiment.
- ☐ Use Postman evidence from Experiment 2 or create a new test request that represents one action from the hosted page. Example: the user clicks Submit Complaint, the frontend sends POST /reports, the backend stores the complaint image path and returns a status response.
- ☐ Draw the complete access workflow in draw.io: Browser/Mobile Client -> Hosted Frontend -> API Endpoint -> Authentication Check -> Backend/Serverless Logic -> Cloud Storage/Database -> Response -> Monitoring/Log. Paste the hosted URL, browser screenshot, workflow diagram and GitHub repository link.

Observe

Verify whether the hosted page or simulated access URL opens, whether the API/storage flow is technically consistent, and whether monitoring/logging evidence is identified.

Record Evidence

Evidence Item	Student Reference / Screenshot Number	Checkpoint
Hosted page URL or approved simulation evidence.		[] Completed
Browser access screenshot.		[] Completed

API/storage workflow diagram.		[] Completed
Monitoring/logging metric table.		[] Completed
GitHub repository link.	https://github.com/SAISANTHOSH192511043/csa1517-cloudcomputing.git	[] Completed

Explain

Explain the complete request flow from client action to cloud output and monitoring.

Student Explanation Space

Student Explanation Space

The complete request flow begins when a user performs an action in the web or mobile application, such as uploading a crop image or requesting a price prediction. The request is sent from the client device to the hosted frontend application. The frontend forwards the request to a secure API endpoint.

The authentication layer verifies the user's identity and permissions before allowing access to backend services. Once authenticated, the backend or serverless function processes the request and interacts with cloud storage or databases to retrieve or store data.

The cloud service performs the required operation, such as storing an uploaded file, running an AI prediction model, or updating records. The processed result is then returned as a JSON response to the frontend application.

Finally, monitoring and logging services record important information such as request count, response time, storage usage, and errors. This helps administrators track system performance, identify issues, and ensure reliable operation of the cloud application.

Extend to Capstone

Extend the workflow to the capstone product carried forward from CO2 AT3, showing how the planned infrastructure now exposes usable cloud services.

Capstone Title	AGRIMARKET AI:CROP PRICE FORECAST AND SELLING DECISION APP
CO2 AT3 Infrastructure Decision Carried Forward	Use a secure API-based architecture between users and cloud databases
Cloud Service / Access Layer Added in this Experiment	Firebase Storage (Object Storage) for storing uploaded files, crop images, reports, and backup data.
Risk or Limitation Identified	APIs without authentication can expose sensitive data and may be vulnerable to misuse or attacks.
Improvement for Prototype or Pilot	Add API authentication, authorization, rate limiting, input validation, and HTTPS encryption for secure access.

Submit

GitHub Folder	CO3_AT1/Experiment_5
Files to Upload	Completed PDF, diagrams, screenshots and any exported table.

Google Classroom Entry

Paste GitHub repository link and attach final PDF.

Final Submission Checklist

Checklist Item	Status
Experiment 1 completed with storage model evidence	☐
Experiment 2 completed with API evidence	☐
Experiment 3 completed with serverless/event-flow evidence	☐
Experiment 4 completed with edge/fog/cloud reasoning	☐
Experiment 5 completed with cloud application access workflow	☐
All diagrams exported and pasted	☐
All screenshots are readable and privacy-safe	☐
Capstone extension completed for every experiment	☐
GitHub folder created and link added	☐
Google Classroom submission completed	☐

Student Assurance

Declaration	I confirm that this CO3 AT1 guided experiment workbook contains my own cloud access evidence, diagrams, observations, explanations and capstone extensions. I have not shared private credentials or sensitive information.
Student Signature	Vallapu Mohan Kumar
Faculty Verification	
Date	